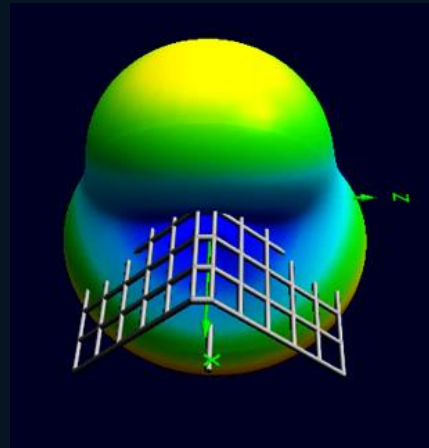
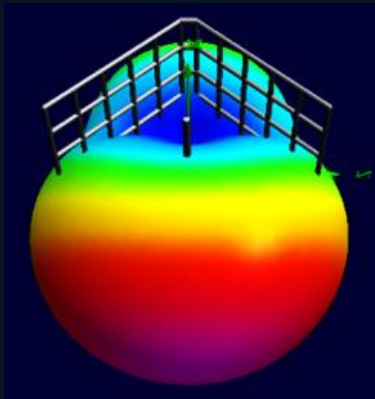
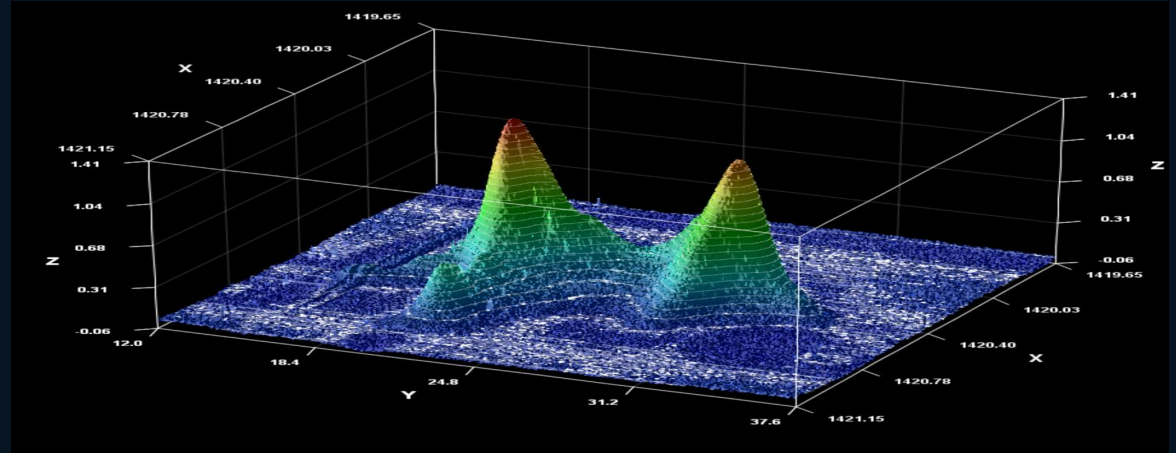
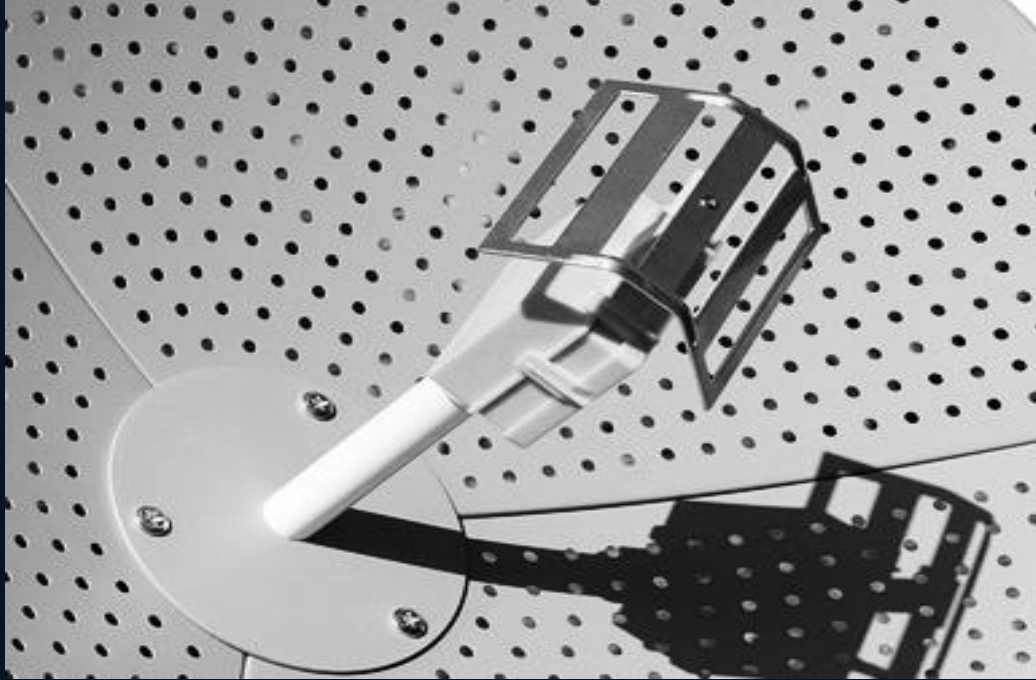
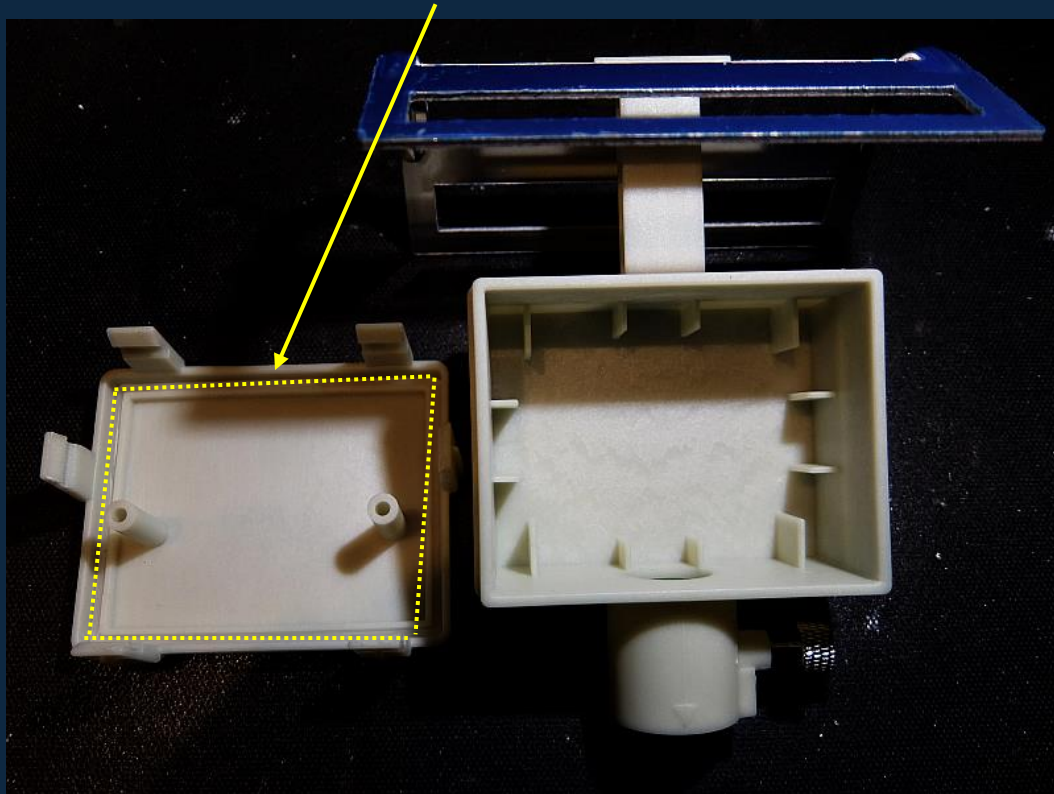


Kraken Discovery Dish 1420 MHz Feed Evaluation using a 1m WiFi Parabolic Reflector

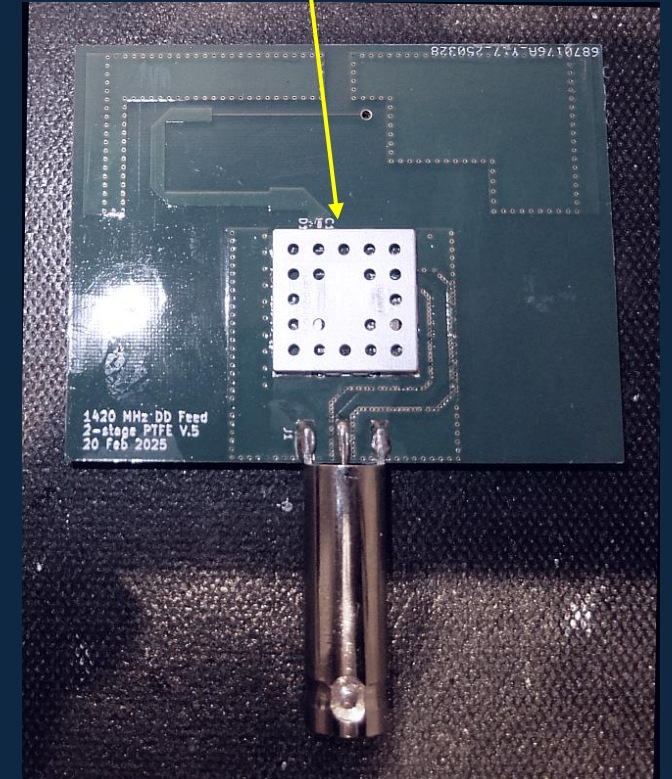
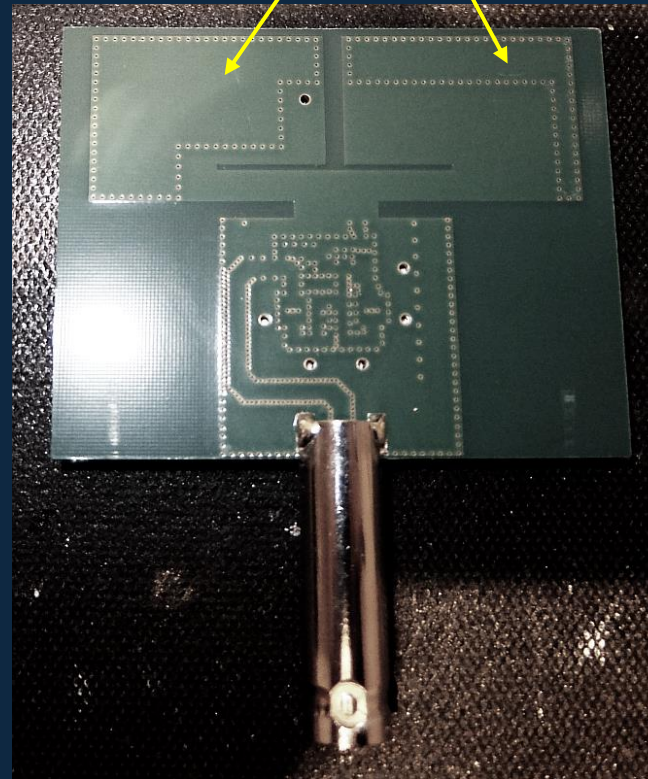


Kraken Discovery Dish 1420 MHz Feed Evaluation Construction

Box has Weather-Seal around perimeter



Antenna is closely coupled to LNA
to Minimize Noise Figure (loss)



Kraken Discovery Dish 1420 MHz Feed Evaluation

LNA



QPL9547

0.1 ~ 6 GHz Ultra Low-Noise Amplifier

Qorvo QPL9547 is optimized for 5G Massive MIMO (Multiple Input Multiple Output) applications. At 1.9GHz, the amplifier typically provides 19.5dB gain, +39dBm OIP3 at a 65mA bias setting, and a noise figure of 0.3dB.

Key Features

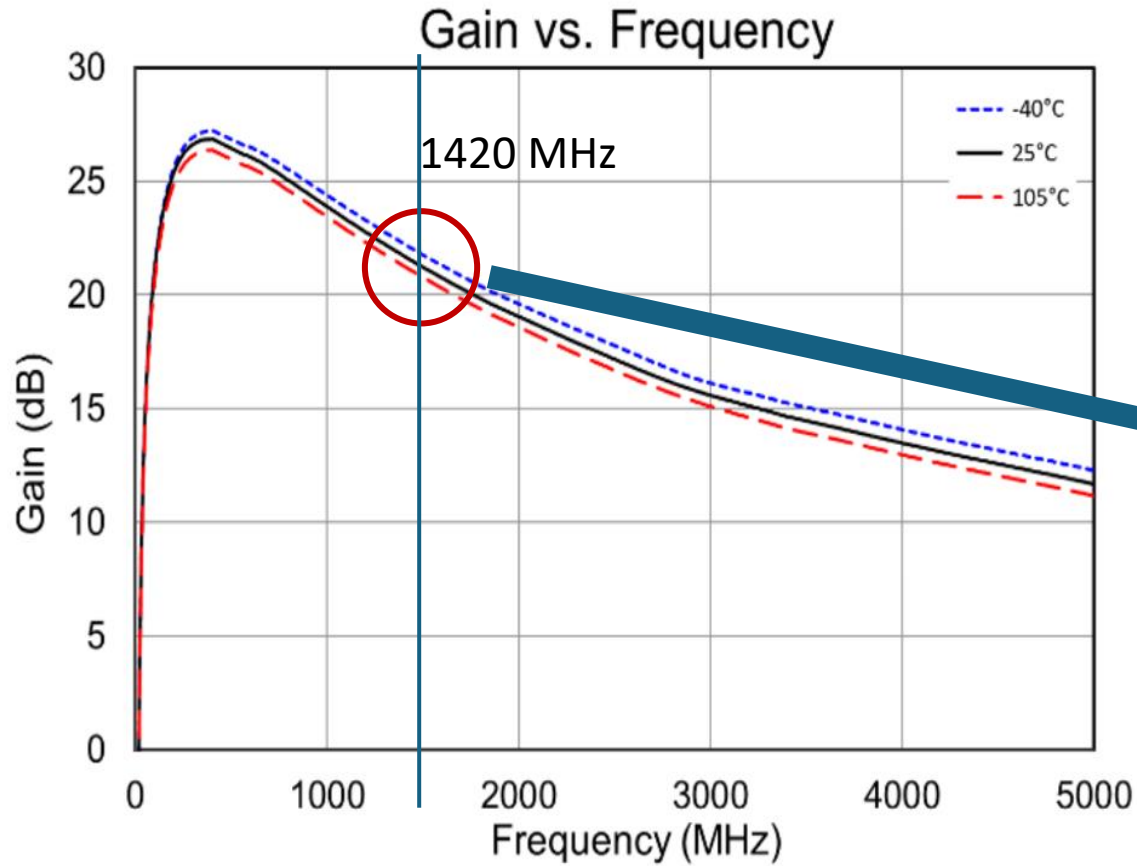
- 0.1-6 GHz Operational Bandwidth
- **Wide bandwidth: operates from 0.1GHz to 6 GHz**
- **Ultra-low noise: 0.3 db NF @ 1.9 GHz**
- **Gain: 19.5 dB typ.**
- **OIP3: +39 dBm typ.**

Qorvo QPL9547 0.1GHz to 6.0GHz Ultra Low-Noise LNA



Kraken Discovery Dish 1420 MHz Feed Evaluation LNA

Qorvo QPL 9547 LNA Information Amplifier Gain vs Temperature Stability (= Very Stable)



Gain vs. Frequency

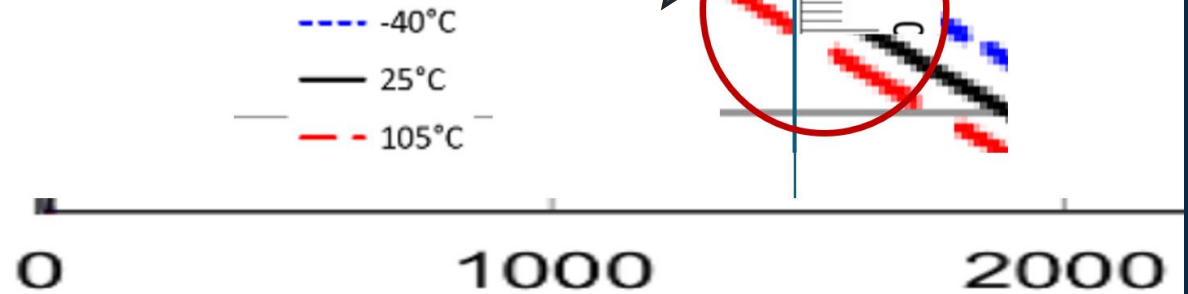
QORVO®

Gain vs Temperature

1.10 dB / 145° C = -0.075 dB / +10° C

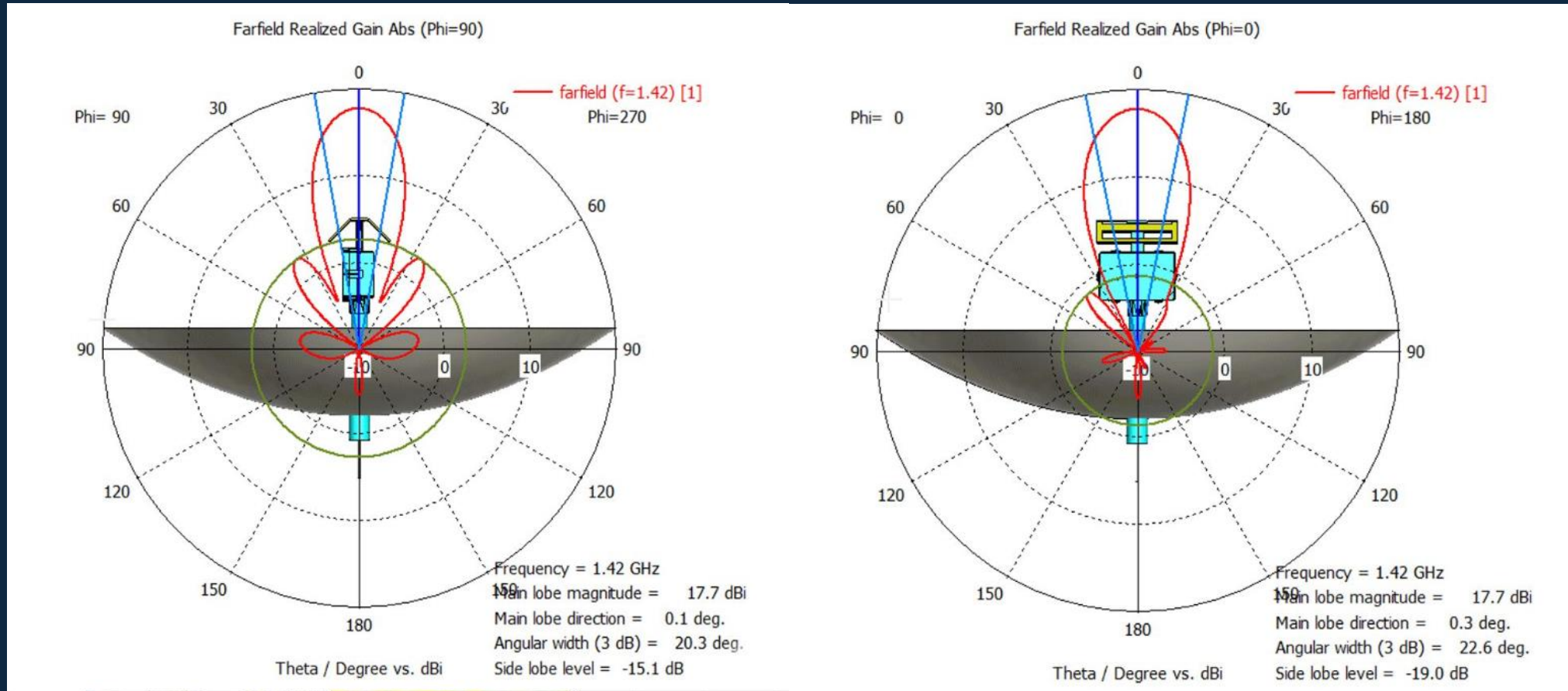
0.45 dB / 80° C = -0.056 dB / +10° C

QPL9547
Ultra Low-Noise Amplifier



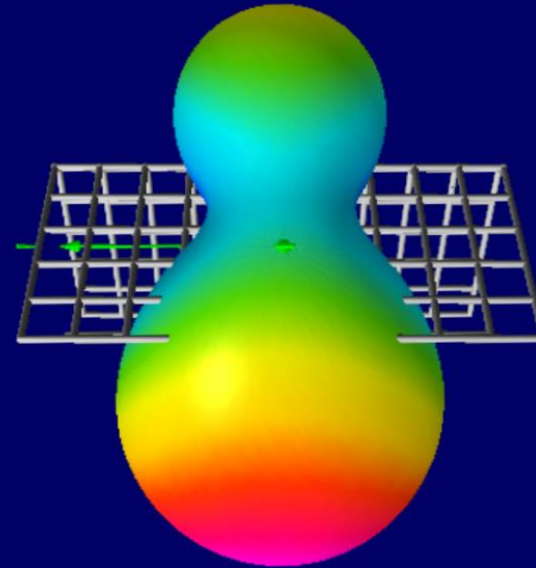
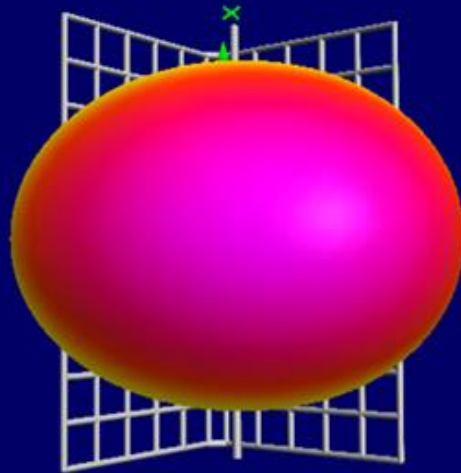
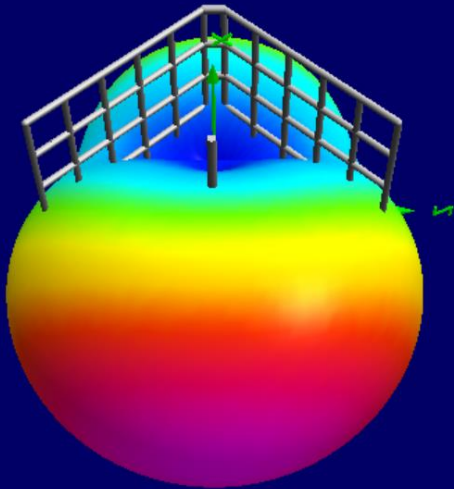
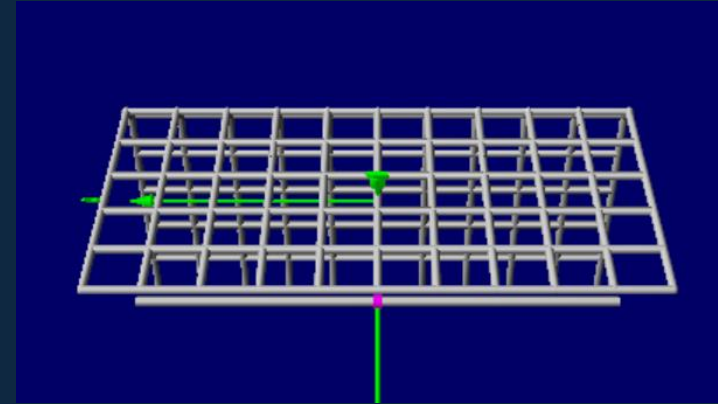
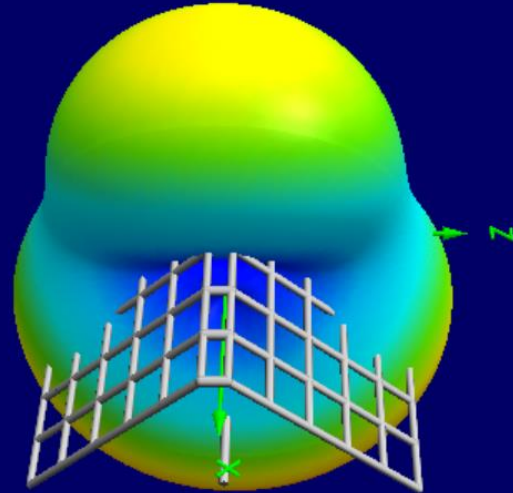
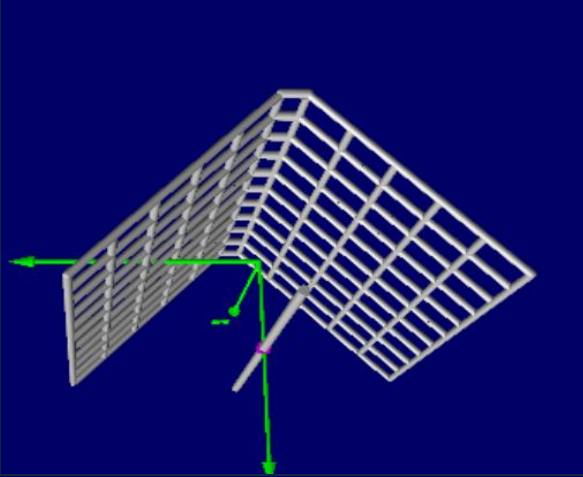
Kraken Discovery Dish 1420 MHz Feed Evaluation

Kraken information : Antenna Beam Patterns



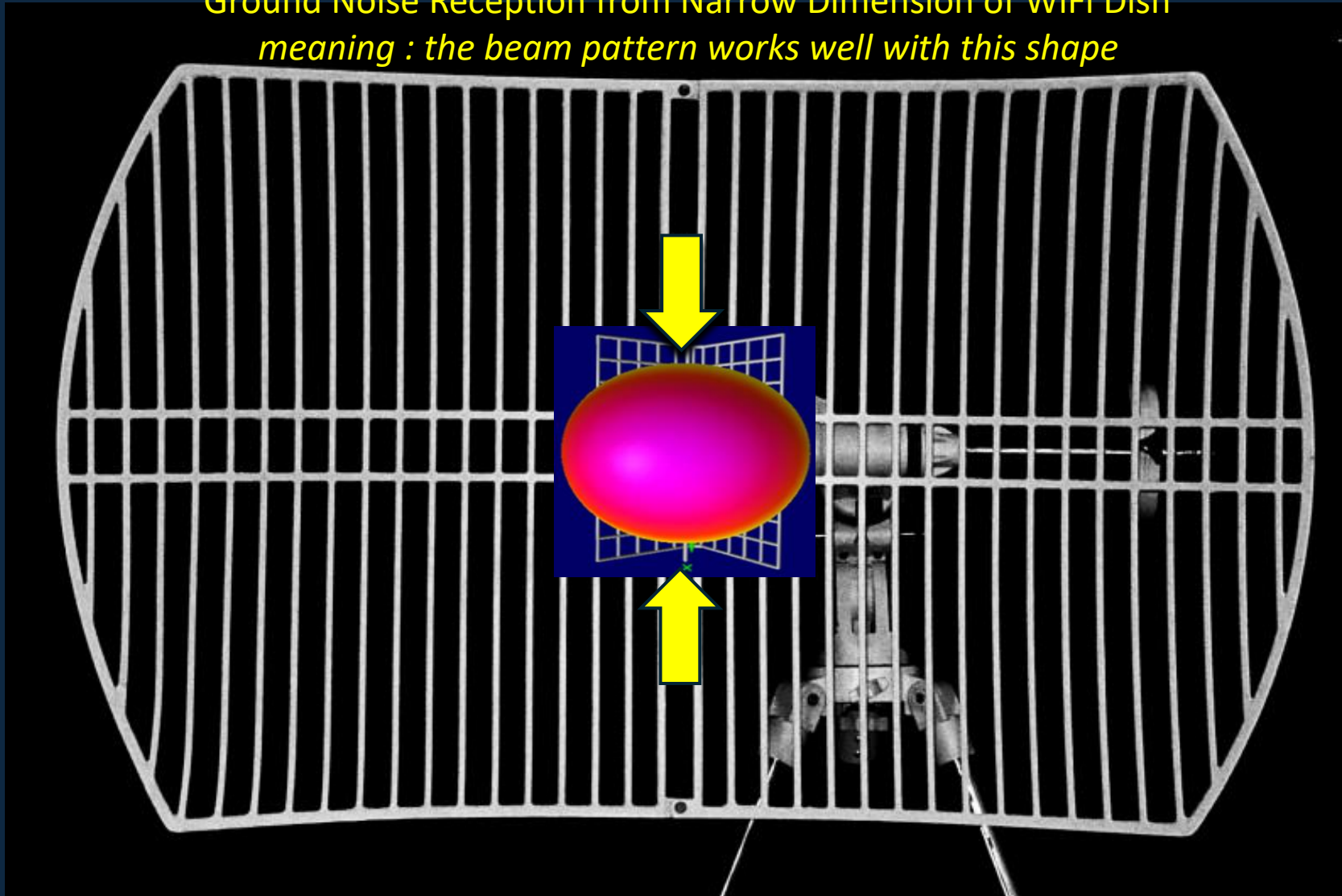
Kraken Discovery Dish 1420 MHz Feed Evaluation
4NEC2 Numerical Electromagnetics Code Dipole/Reflector Analysis
The actual antenna is more complex than this simple dipole .

by alex petitit



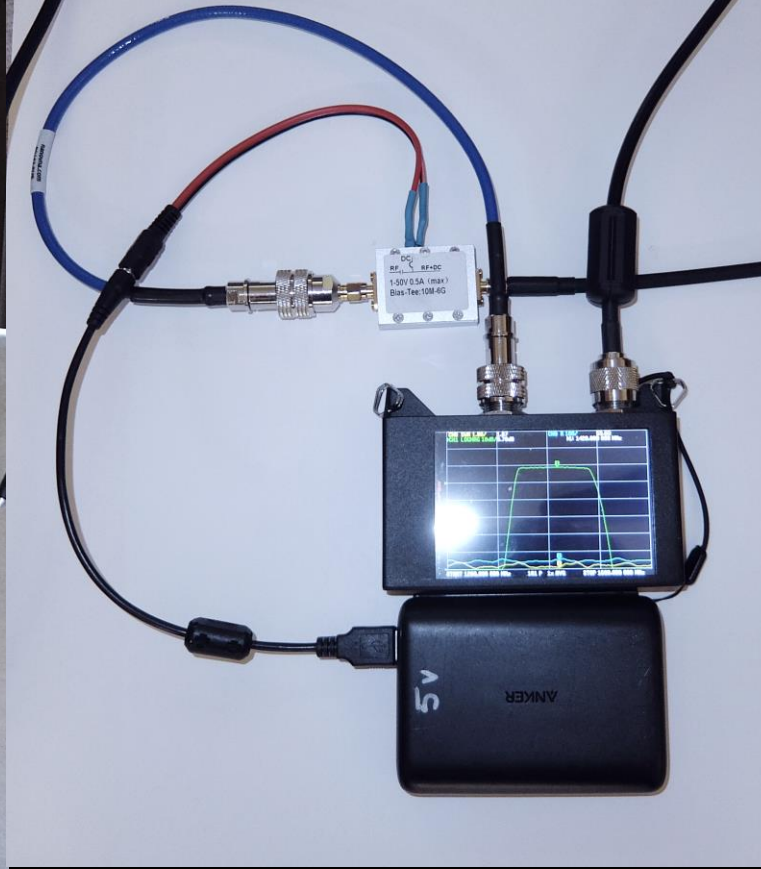
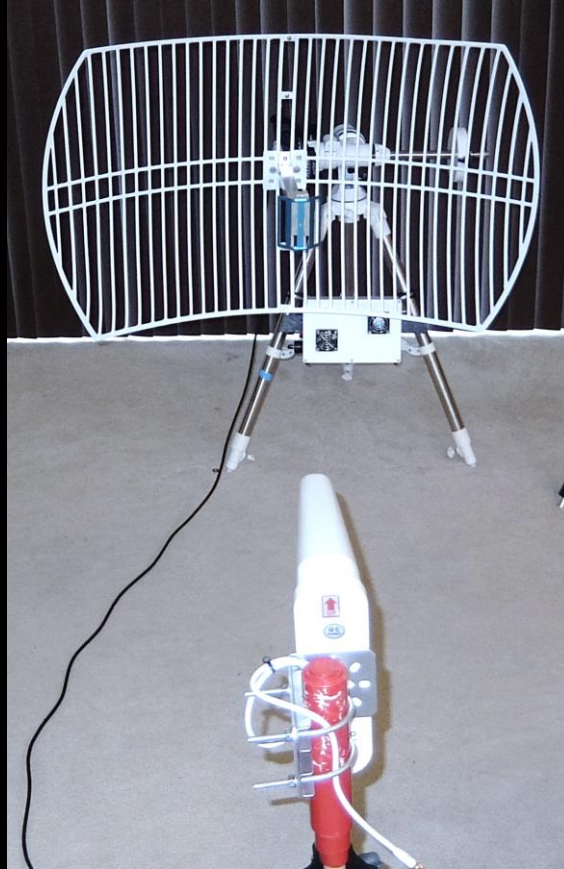
Kraken Discovery Dish 1420 MHz Feed Evaluation
4NEC2 Numerical Electromagnetics Code Dipole/Reflector Analysis
The actual antenna is more complex than this simple dipole .

Narrow Out-of-Axis Beam Pattern Minimizes
Ground Noise Reception from Narrow Dimension of WiFi Dish
meaning : the beam pattern works well with this shape



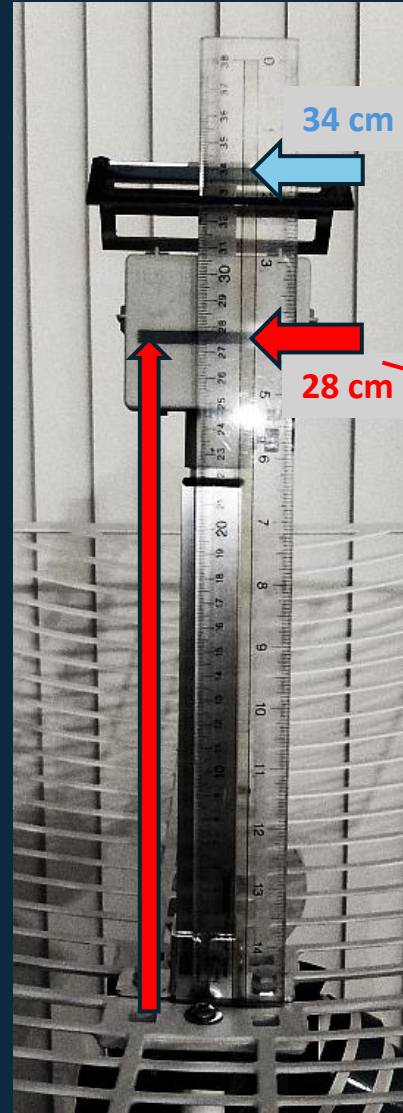
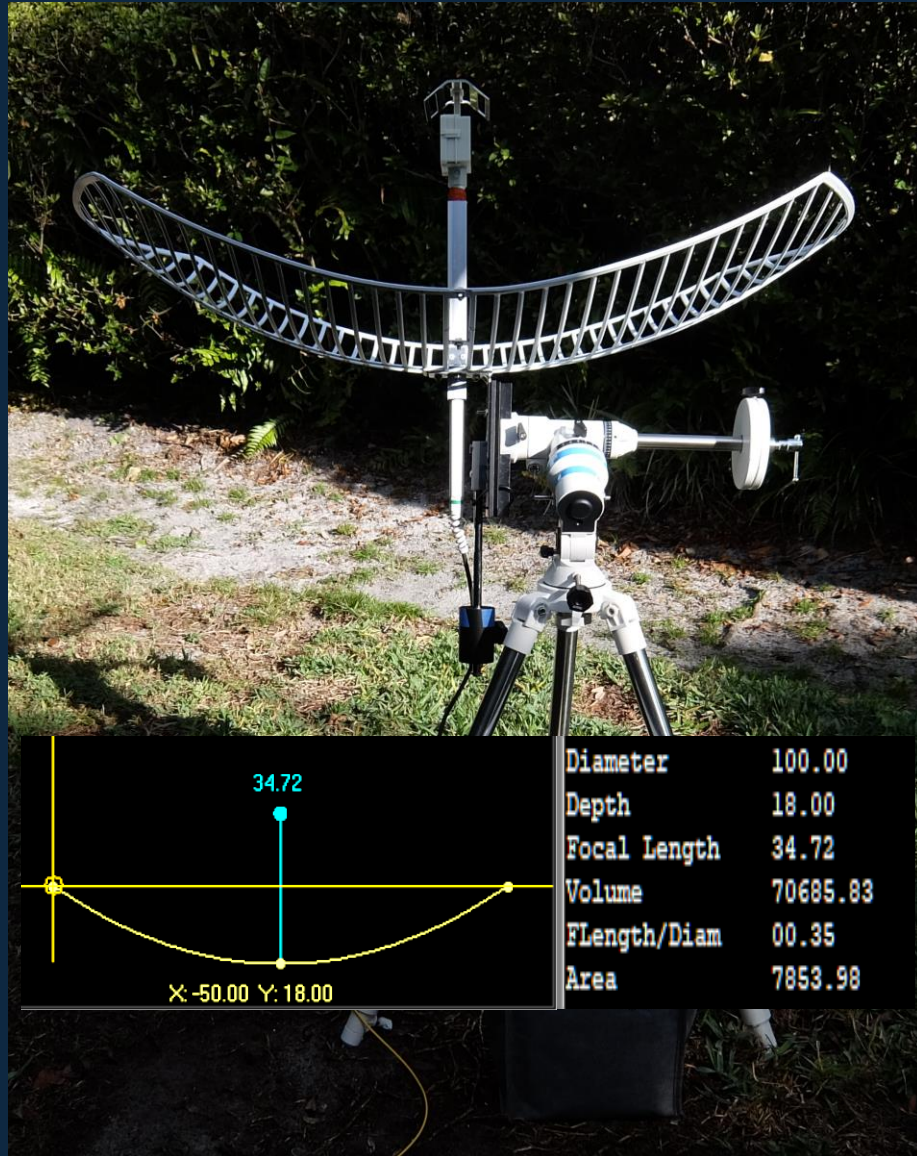
Kraken Discovery Dish 1420 MHz Feed Evaluation

Band Pass Filter Characteristics : 1380 – 1490 MHz

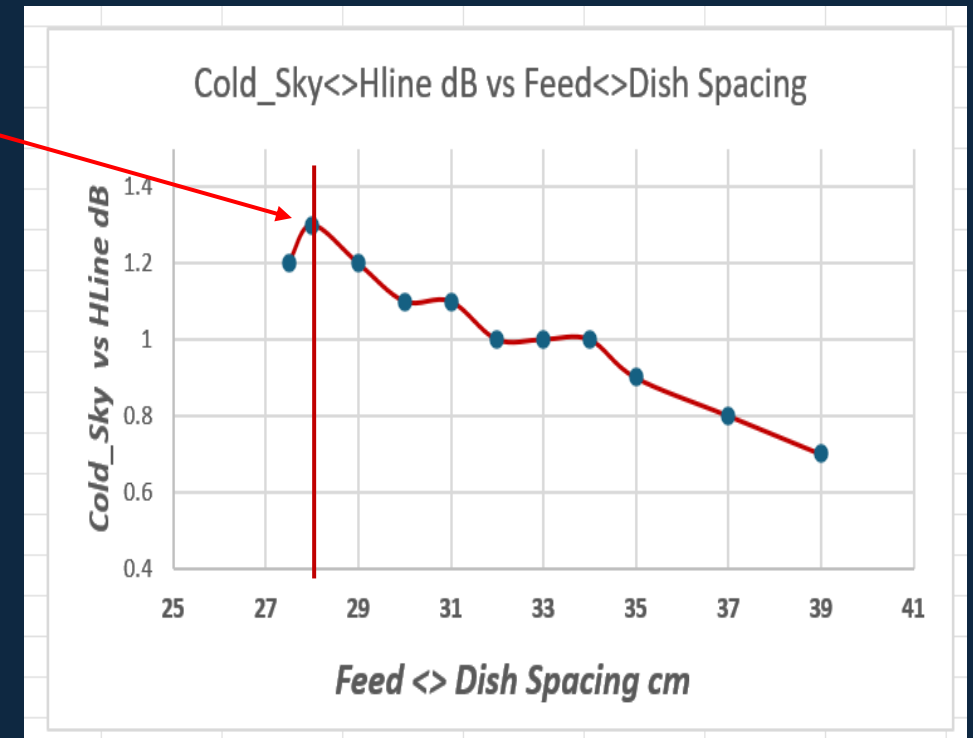


Kraken Discovery Dish 1420 MHz Feed Evaluation

WiFi Dish<>Feed Spacing Optimization



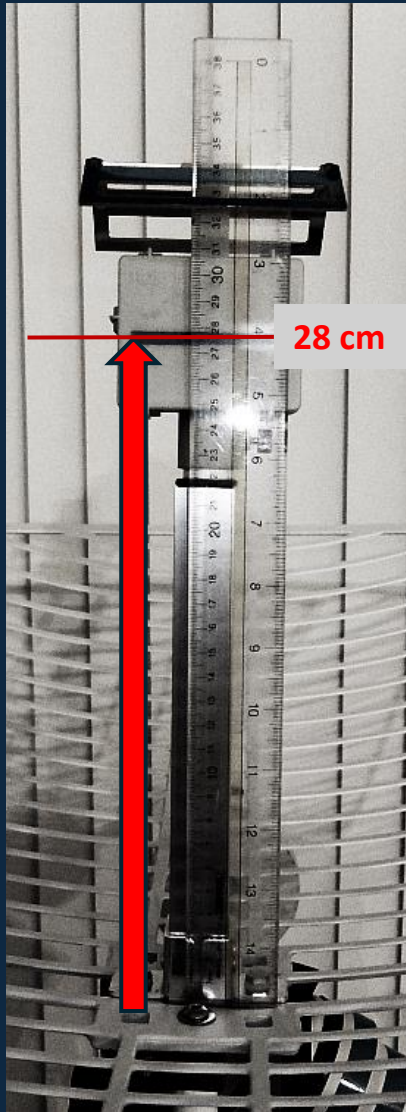
Kraken Discovery Feed 1420 MHz Dipole
Kraken Discovery Feed H1 Internal LNA
Nooelec 1m WiFi Grid Dish



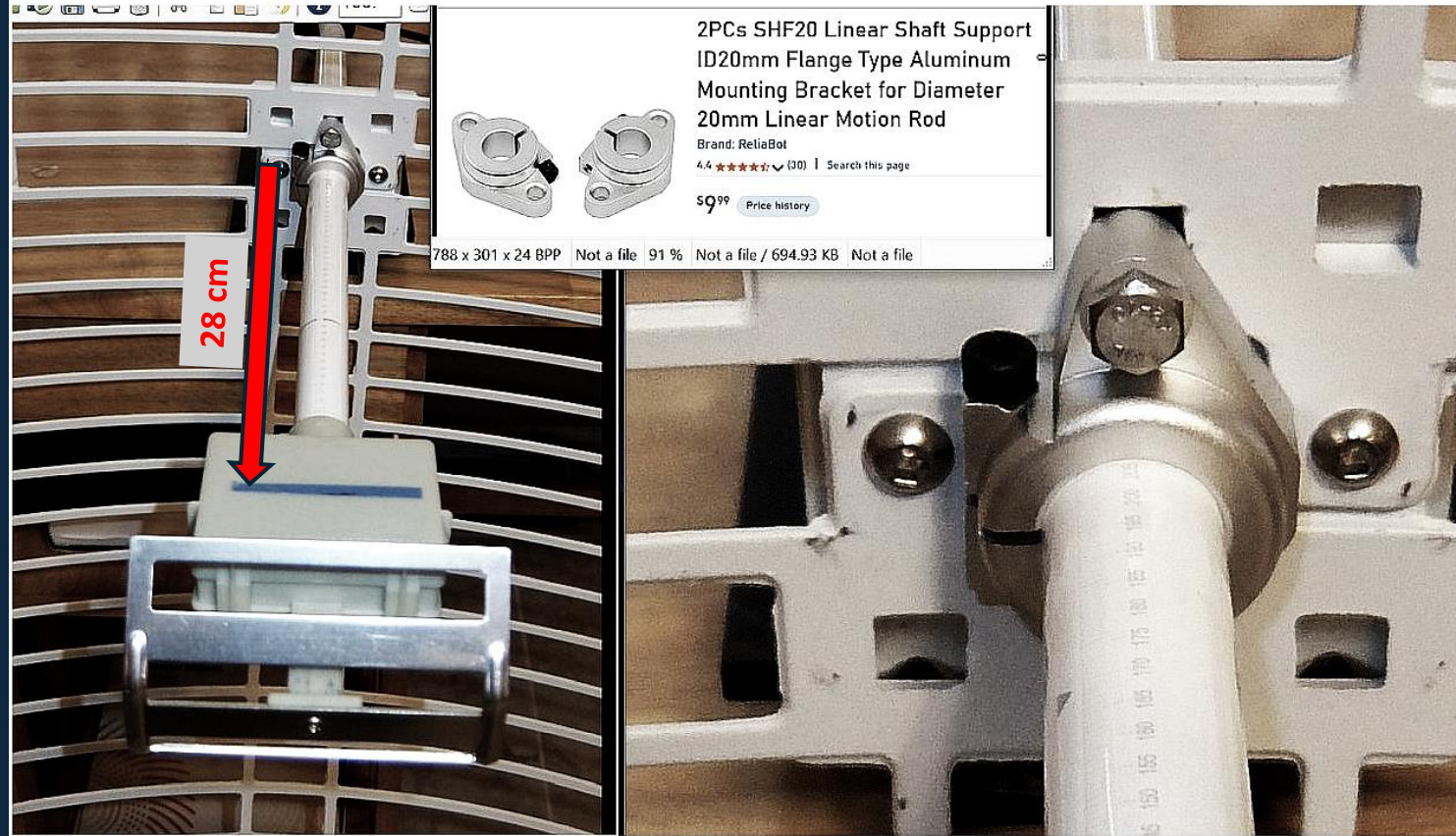
Kraken Discovery Dish 1420 MHz Feed Evaluation

WiFi Dish<>Feed Attachment Flange

SHF20 Linear Shaft Support
ID 20mm Flange Type
Aluminum Mountng Bracket for
20mm Diameter Linear Motion Rod



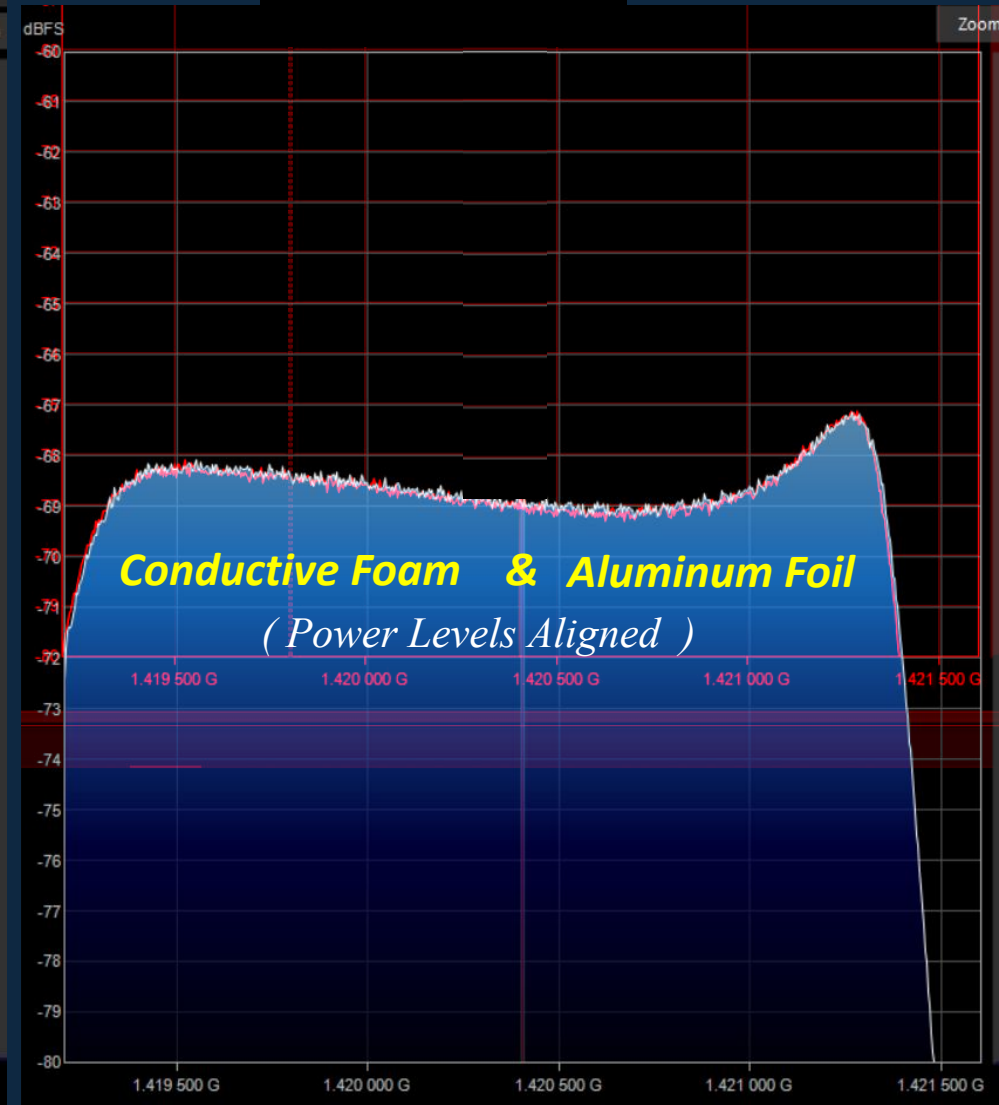
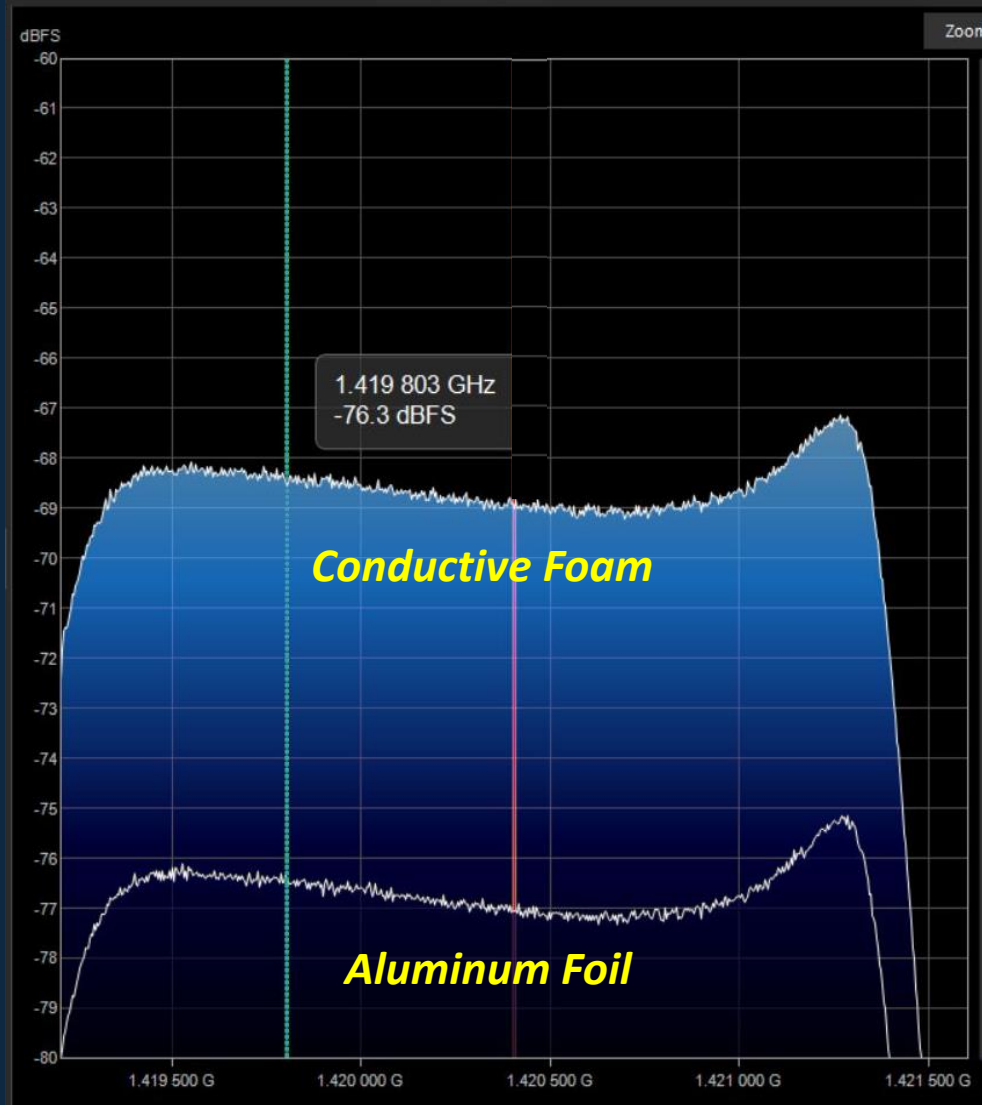
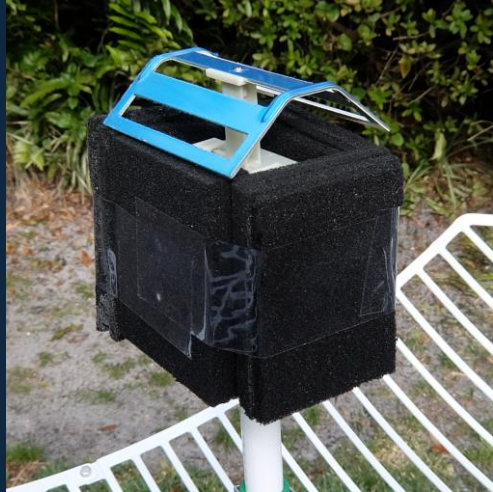
Its a perfect fit between the Flange Adapter & Feed Support Tube



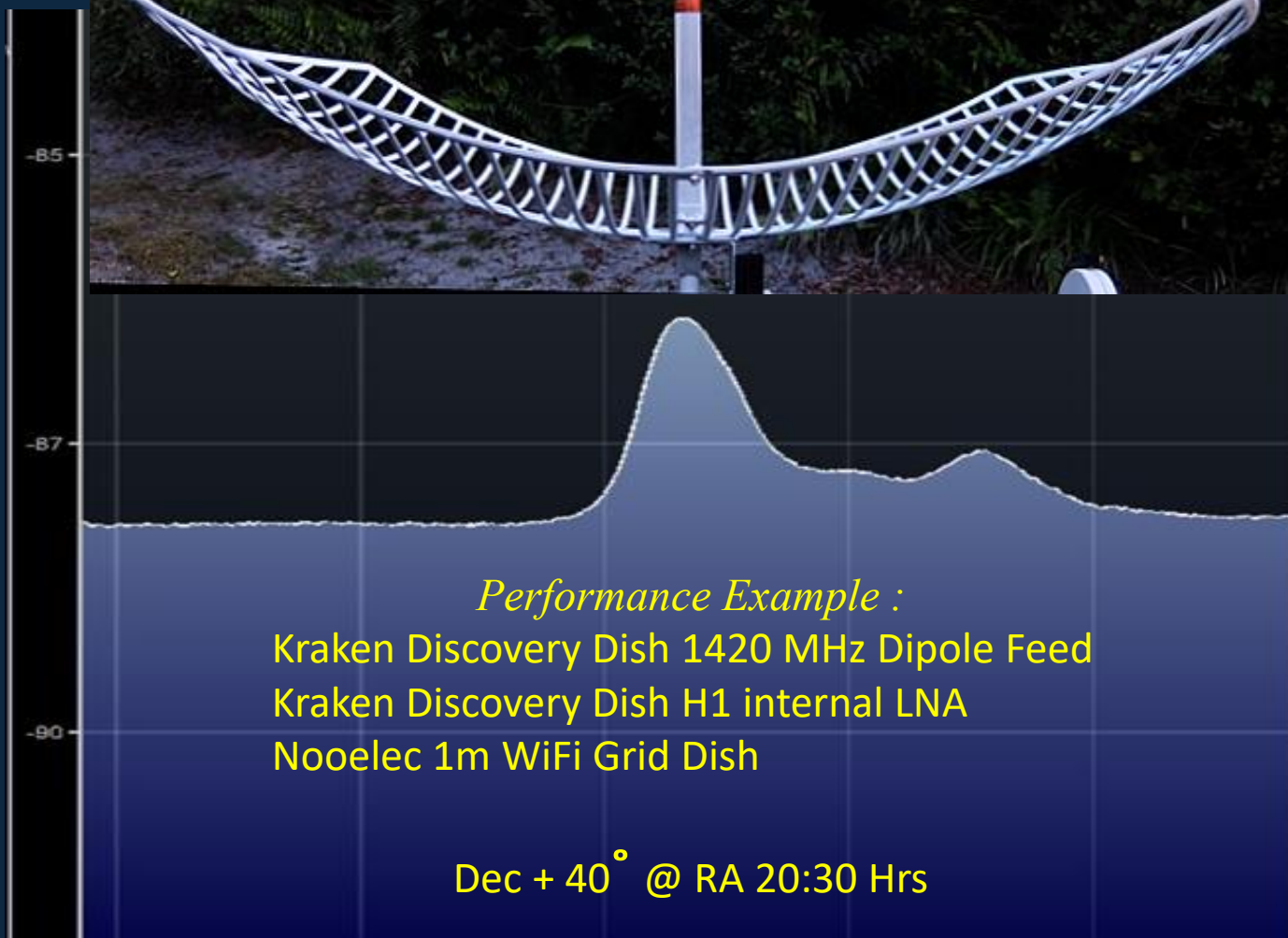
Kraken Discovery Dish 1420 MHz Feed Evaluation

IF_avg Background Correction

Materials Evaluation : *Aluminum Foil works for this application (Conductive Foam Box has a bit less noise)*



Performance Example



Kraken Discovery Dish 1420 MHz Dipole Feed

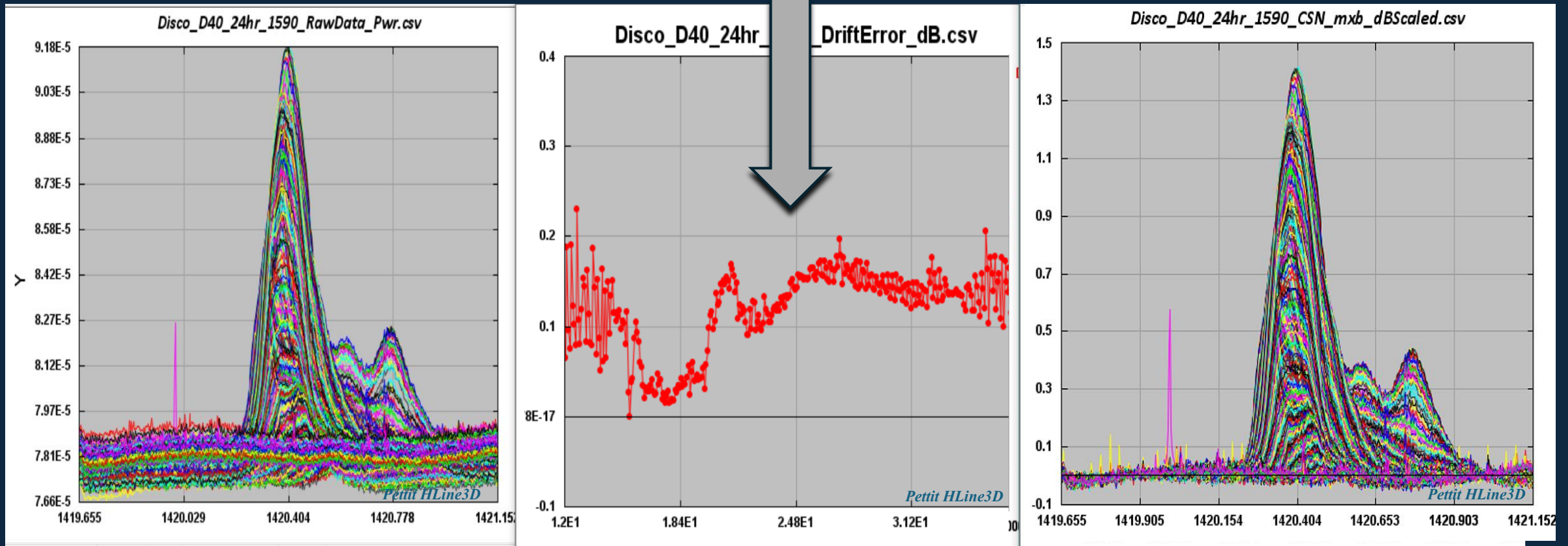
Kraken Discovery Dish H1 internal LNA

Nooelec 1m WiFi Grid Dish

Dec + 40° @ 24 hour Drift Scan

Good Combination : High Quality Data & Low Frame-Frame Drift Error

(This data shows exceptionally low Drift Error)



Kraken Discovery Dish 1420 MHz Dipole Feed

Kraken Discovery Dish H1 internal LNA

Nooelec 1m WiFi Grid Dish

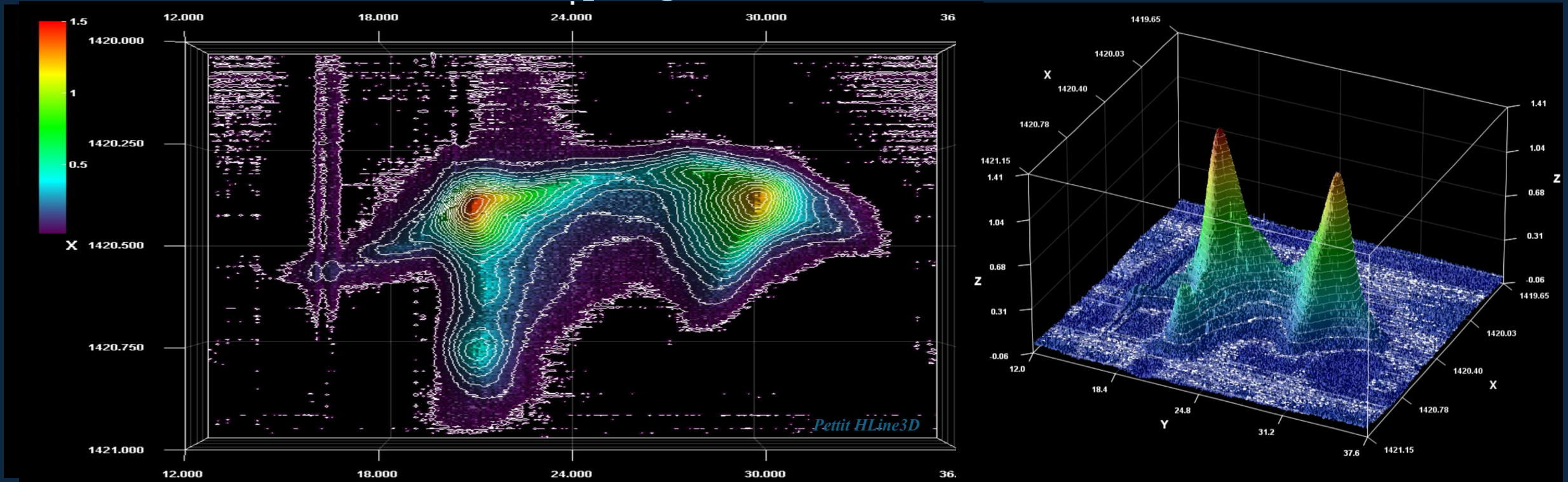
Dec + 40° 24 hour Drift Scan

Excellent **HIGH SIGNAL** / LOW NOISE Performance

Radio Telescope Performance Evaluated Using :

AirSpy SDR# Studio > Kaminski IF_average > Pettit HLine3D > Rinearn Graphics

<https://github.com/AP-HLine-3D/HLine3D>

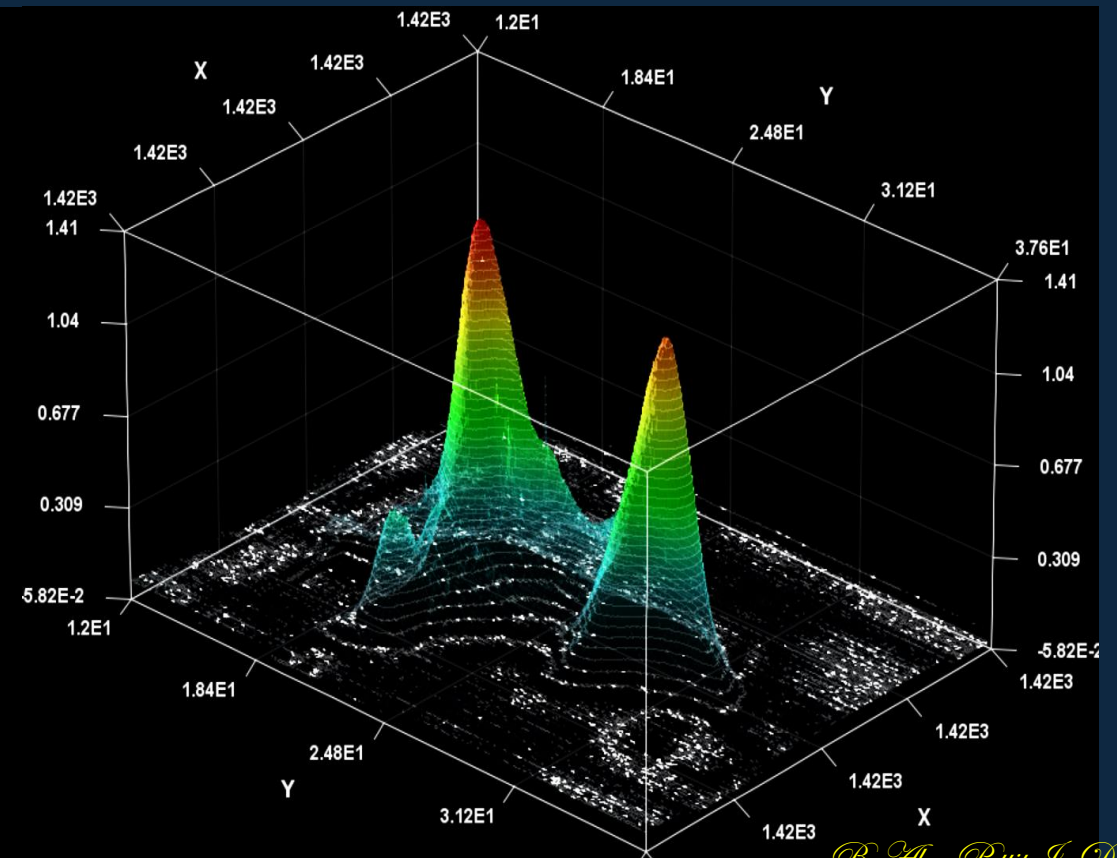
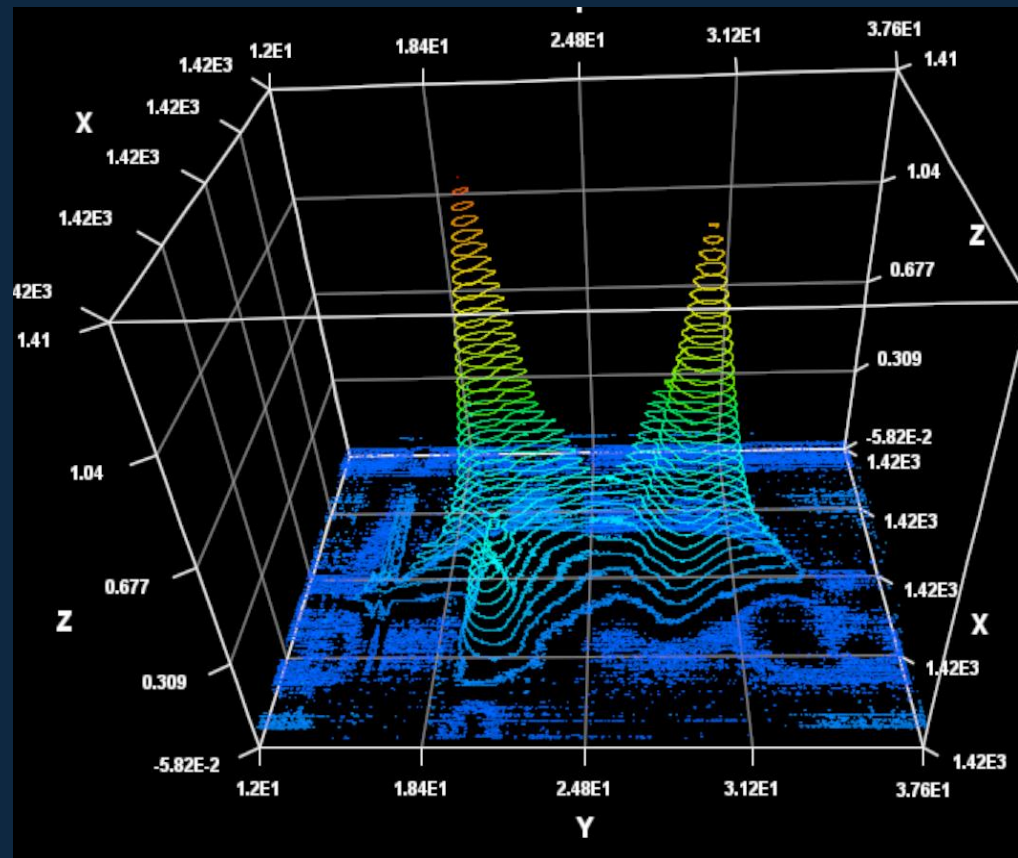


Kraken Discovery Dish 1420 MHz Dipole Feed

Kraken Discovery Dish H1 internal LNA

Nooelec 1m WiFi Grid Dish

EXEC SUMMARY : This is a " Near Off The Shelf "
Hydrogen Line Radio Telescope with Outstanding Performance

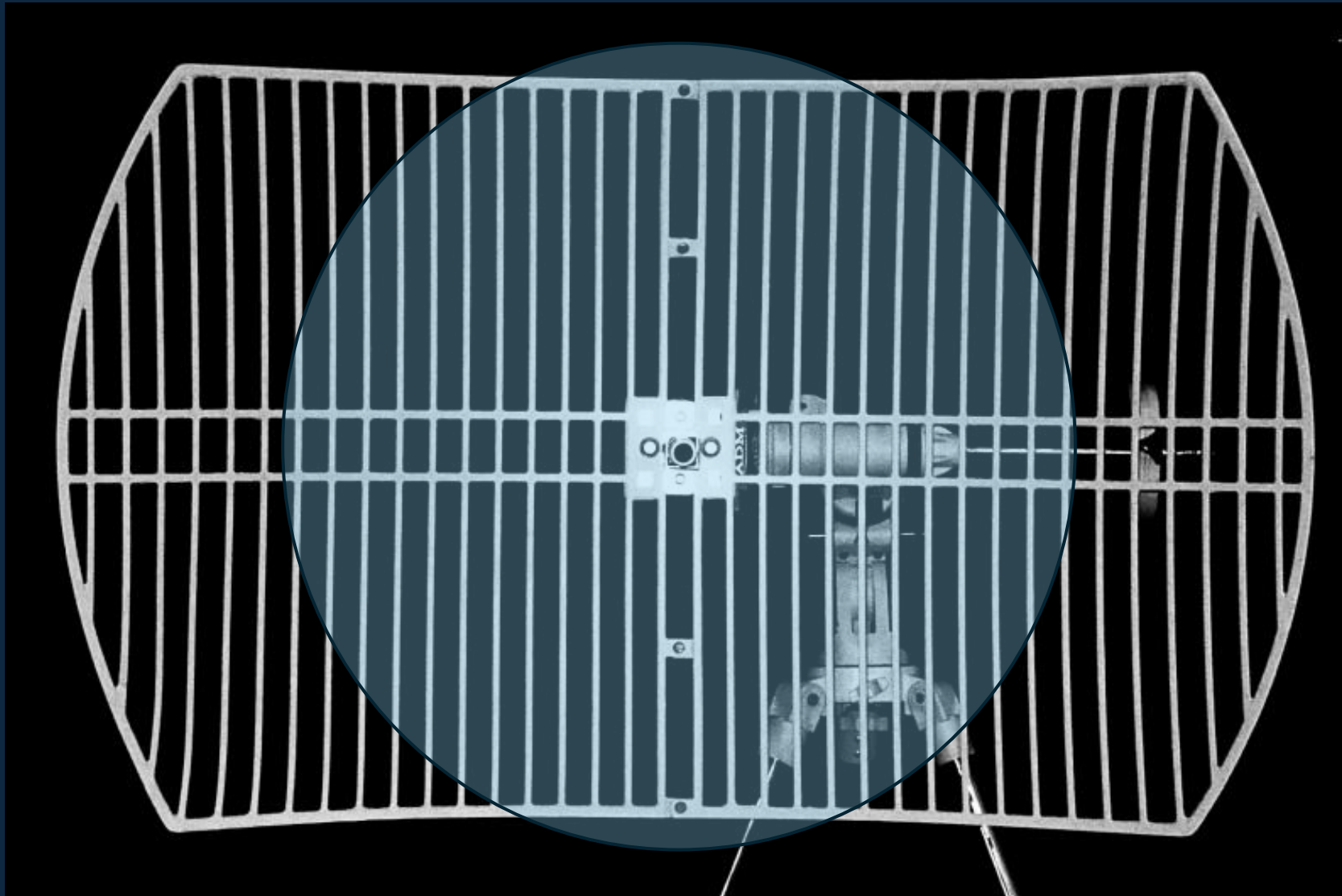


Kraken Discovery Feed 1420 MHz Dish Options

1m x 0.6m WiFi Dish used for This evaluation

0.65m Kraken Discovery Dish will be eval'd in the future

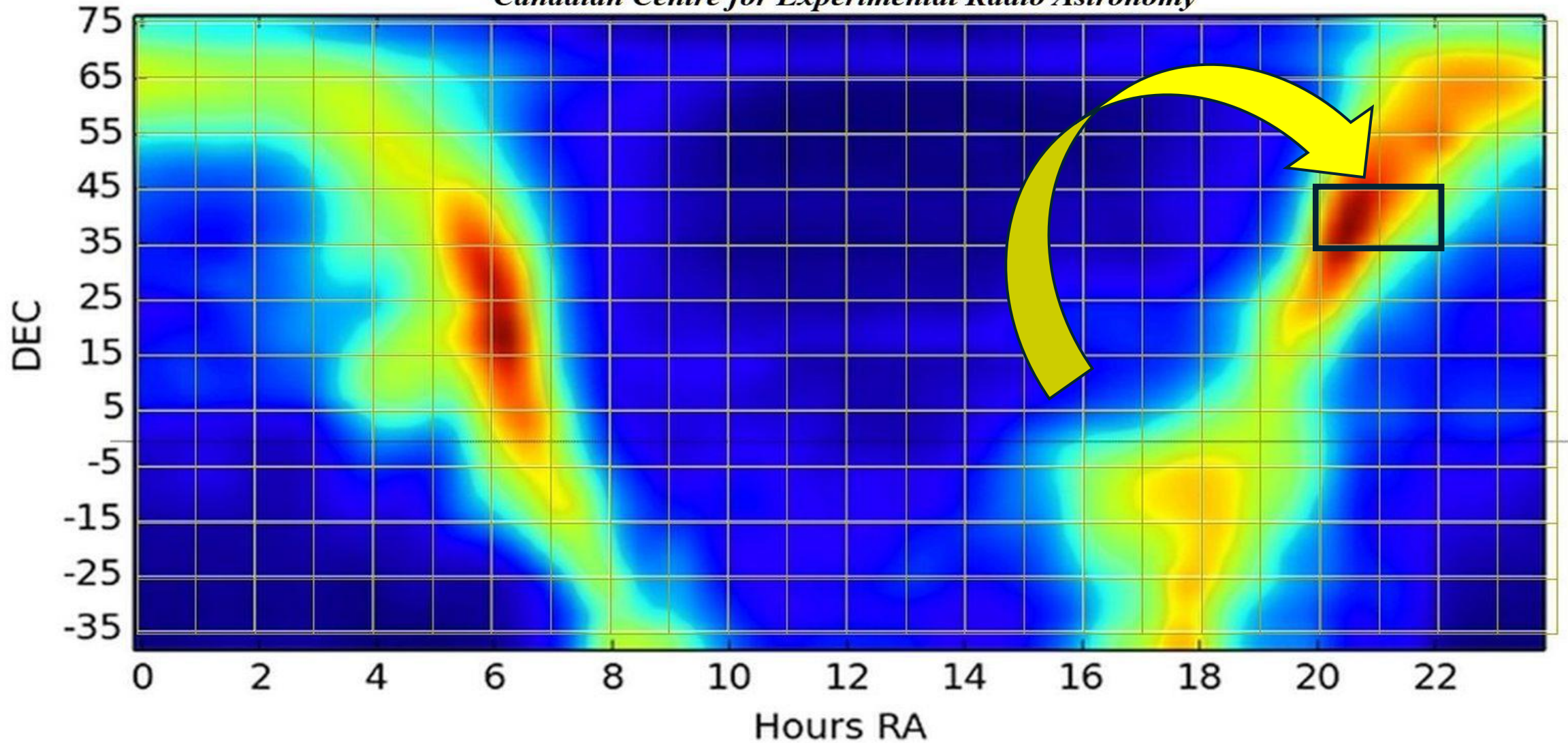
(relative area comparison)



Drift Scan 2 Hr Scan @ Declination + 40° RA 20 - 22 Hrs

CCERA

Canadian Centre for Experimental Radio Astronomy



Drift Scan 2 Hr Scan @ Declination +40° RA 20 - 22 Hrs

Disco_D40_2hrs_CSN_mxb_dBScaled.csv

